

Theory of dynamical stability in attractor neural networks predicts CA3 neurons' activation functions

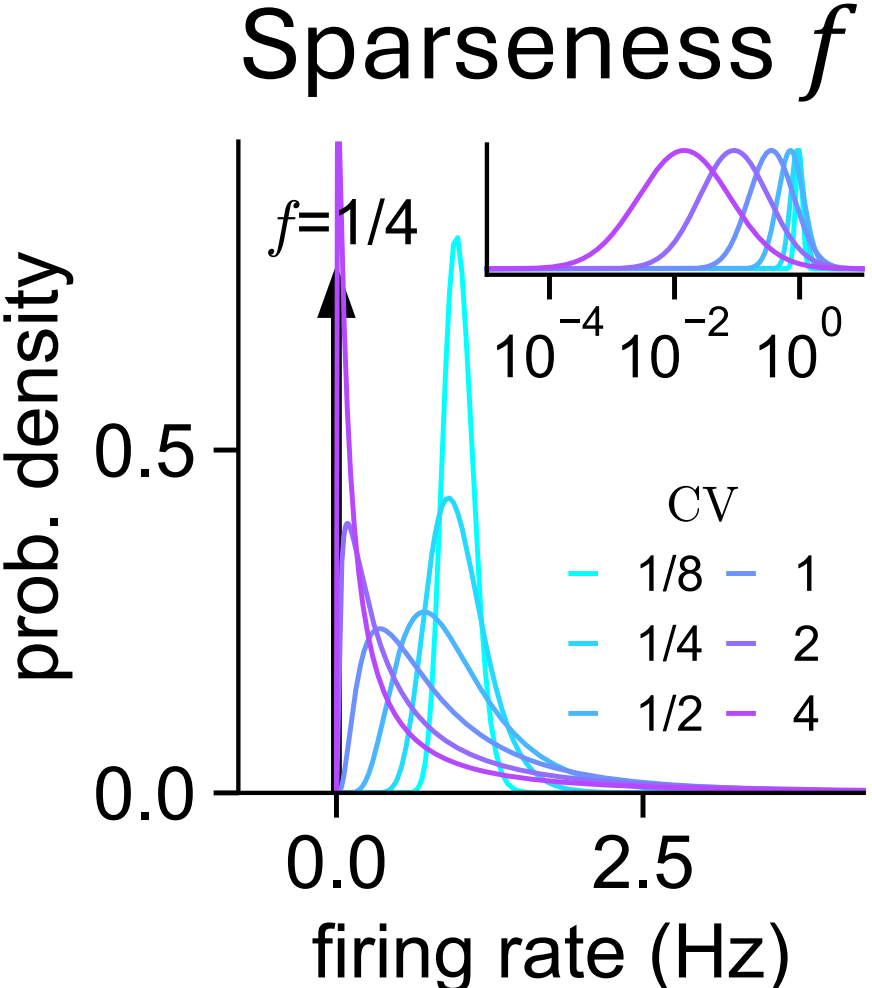
question

Under what conditions can you store rate-coded memory patterns as stable fixed points of a recurrent neural network?

authors

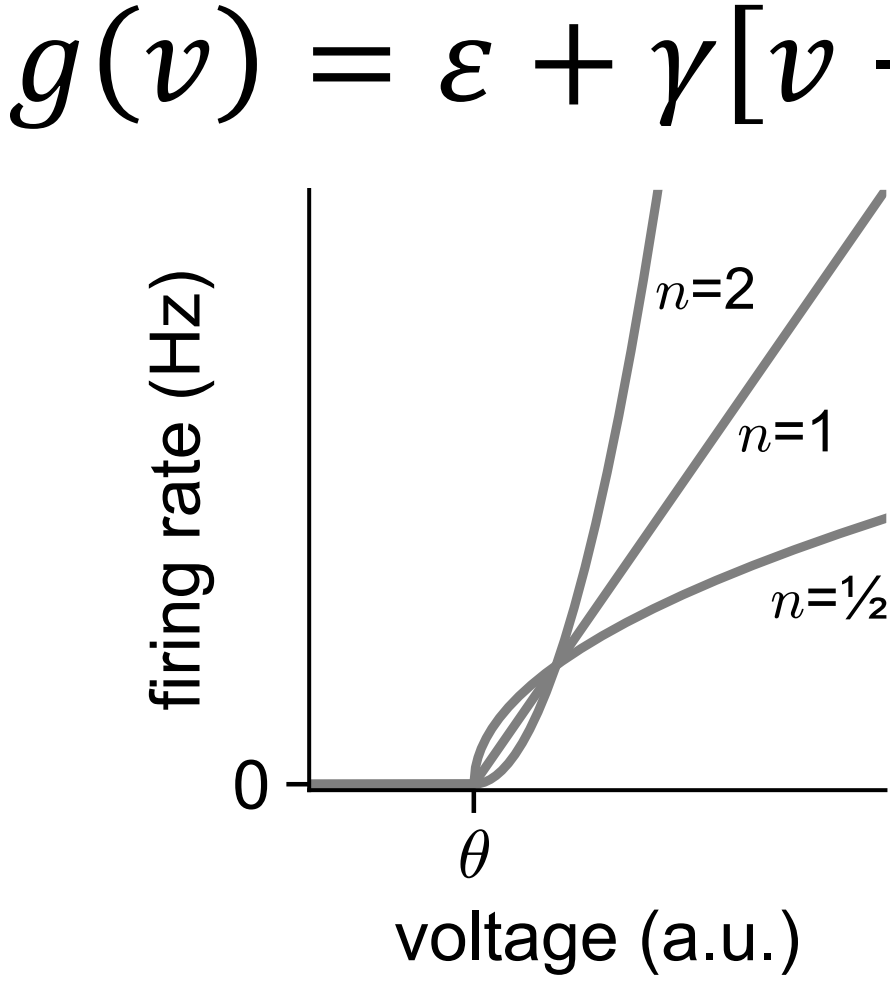
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model

Sparseness f


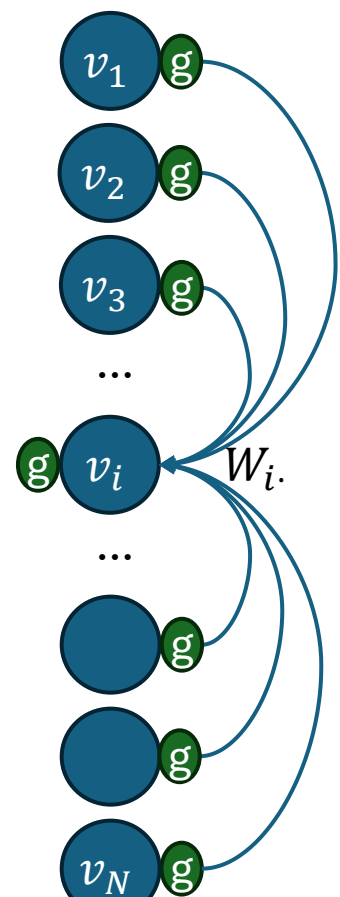
Log-normal firing rate mean 1Hz, different CV

Graded patterns

$g(v) = \varepsilon + \gamma[v - \theta]^n_+$


Neural activation

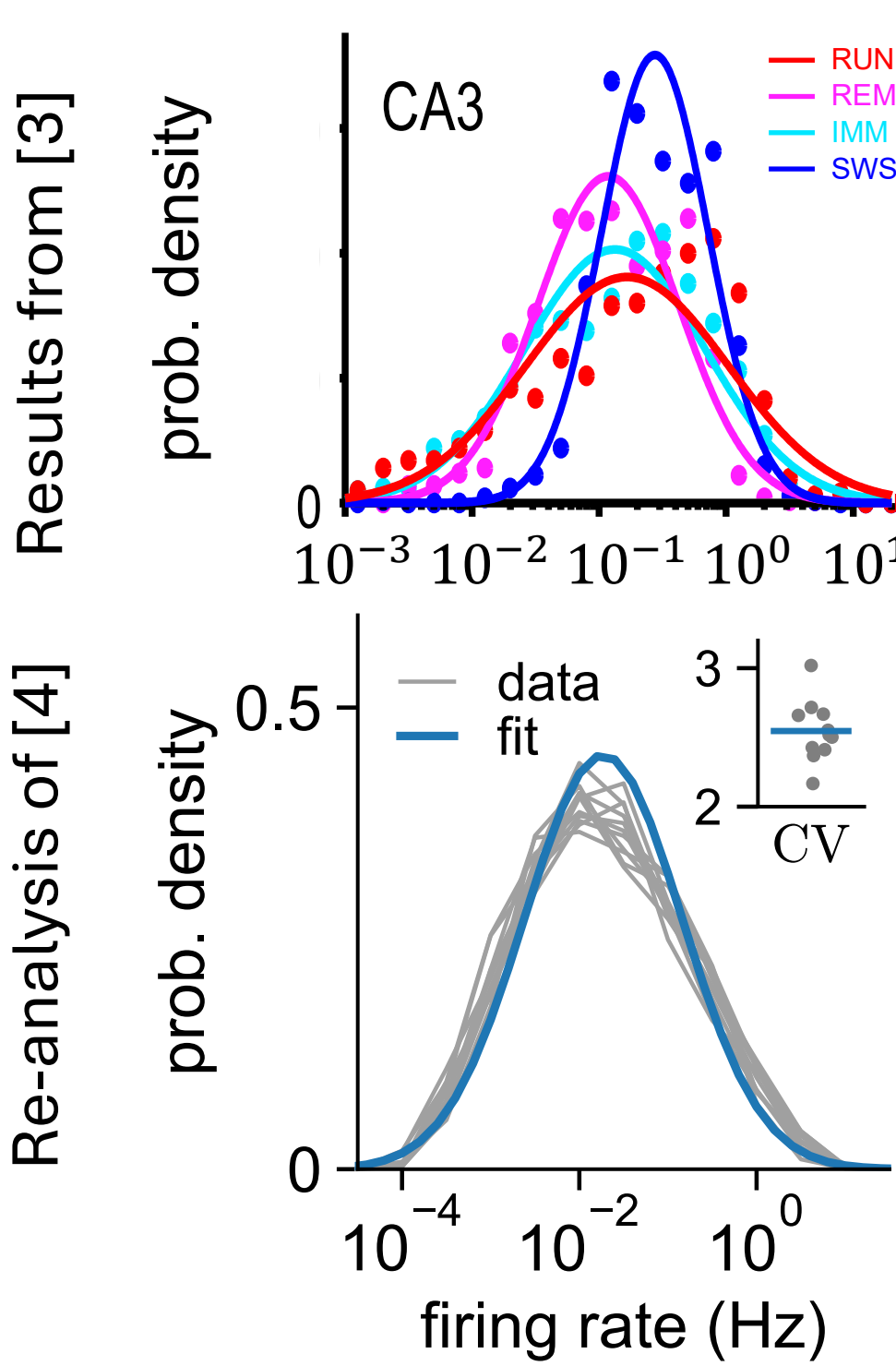
Voltage dynamics
 $\tau \dot{v} = -v + Wg(v)$
 Rate dynamics
 $\tau \dot{r} = -r + g(Wr)$
 Recurrent networks



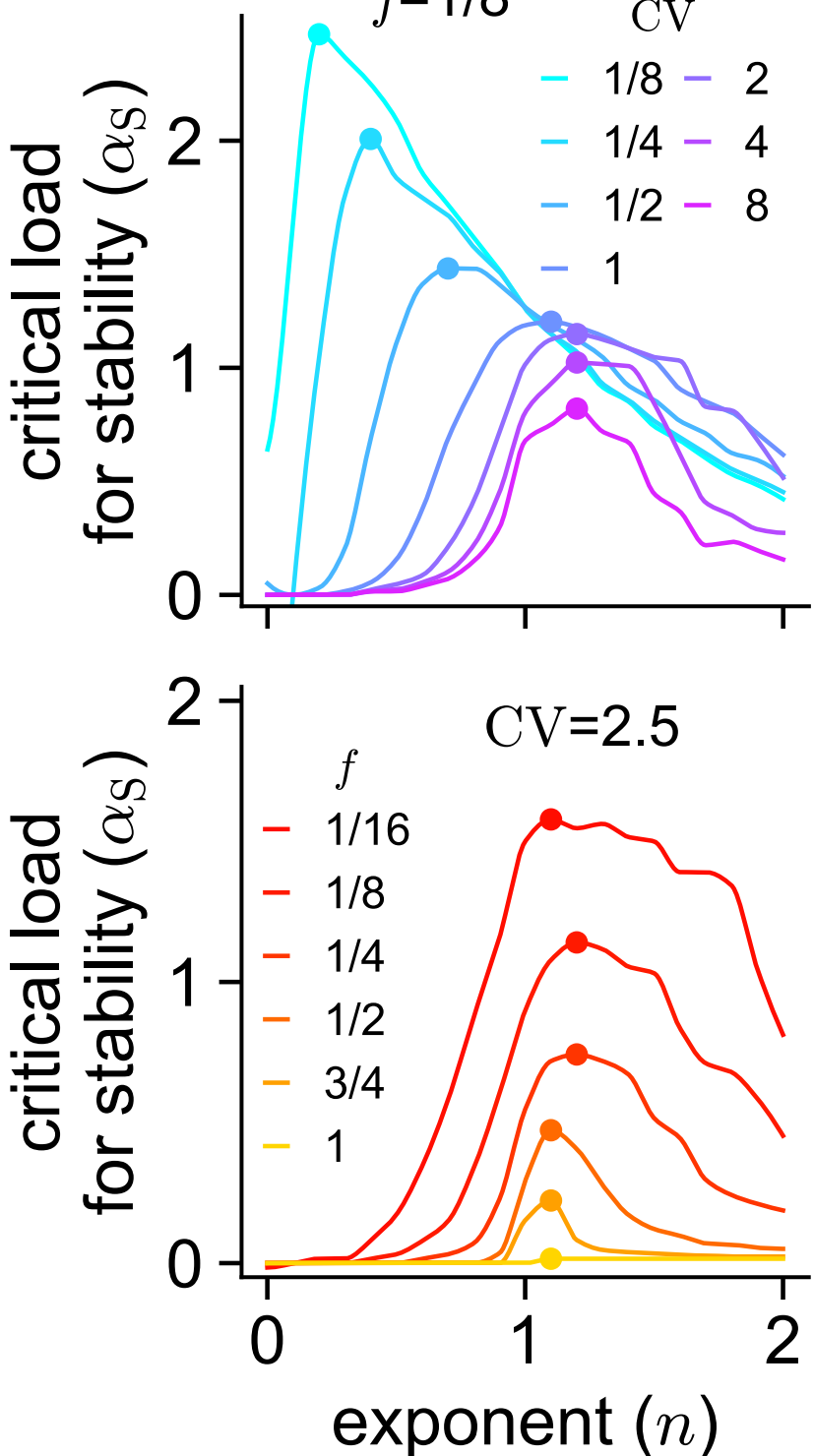
$W^* = \arg \min_W |W|$
 $s.t. \mathbf{r}^\mu = g(W\mathbf{r}^\mu)$
 $\forall \mu = 1 \dots P$

Storing patterns as network fixed points

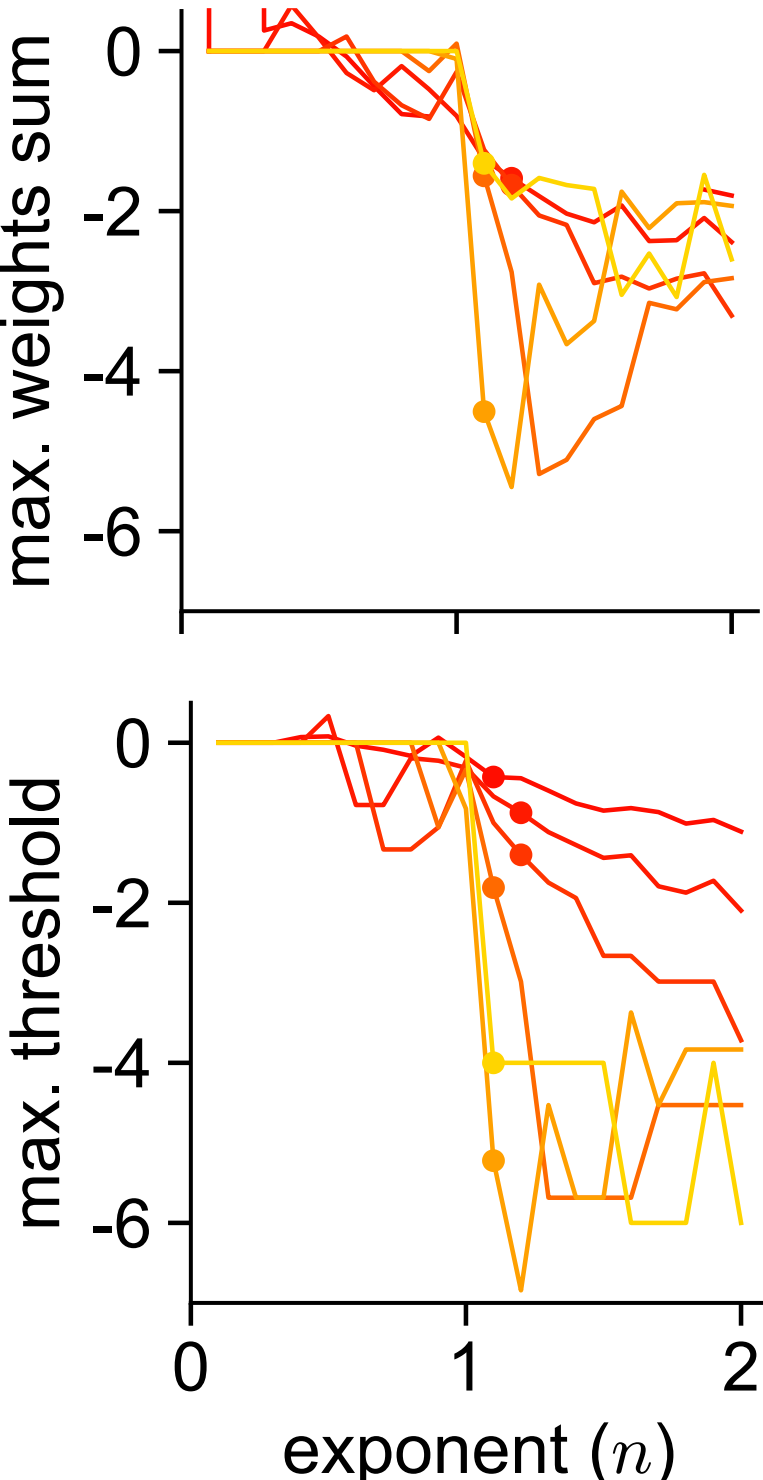
theory



CA3 activity CV is 2-3 high-variance patterns

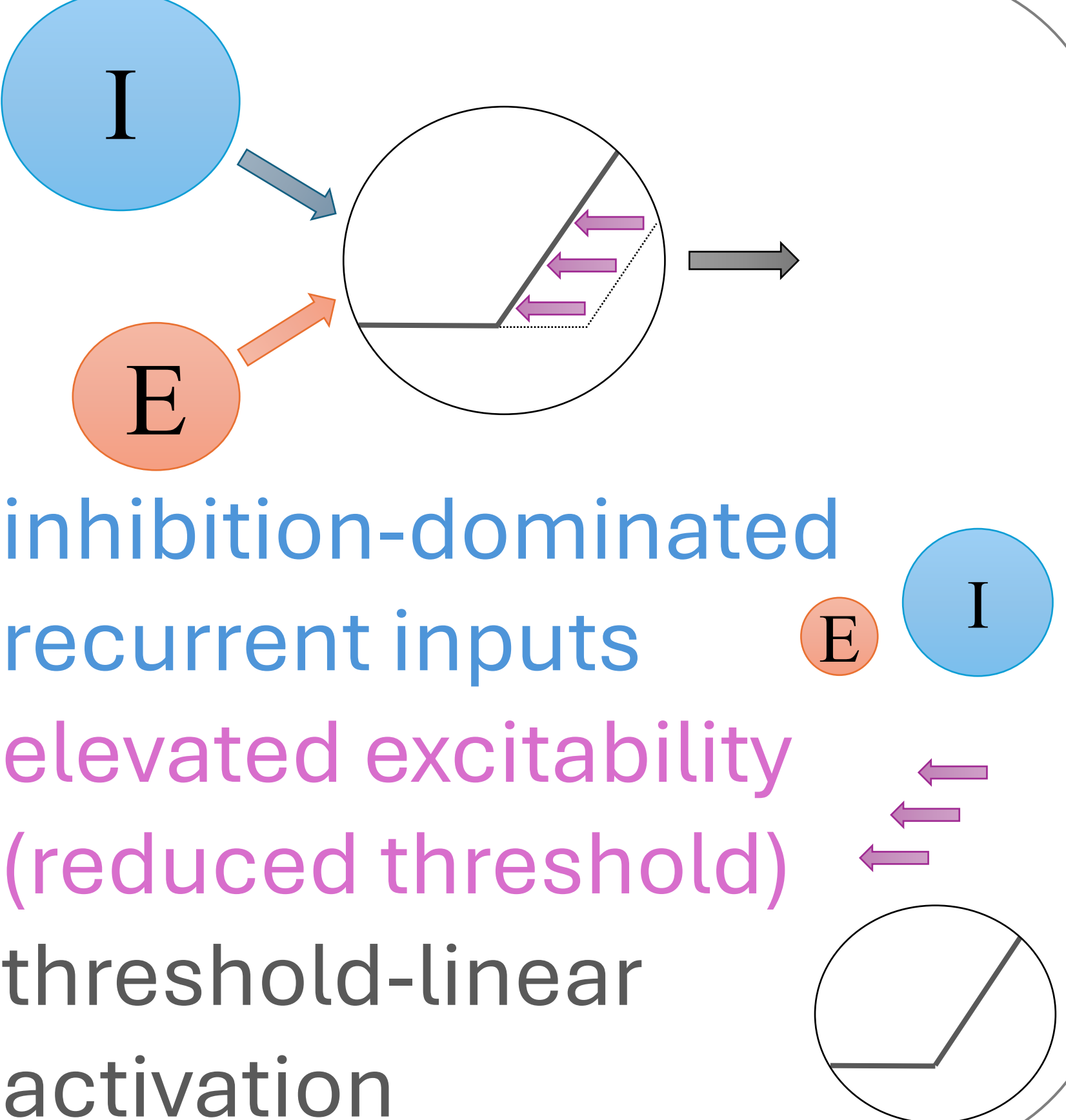


For CV ≥ 1, n^* is 1-1.2 threshold-linear activation



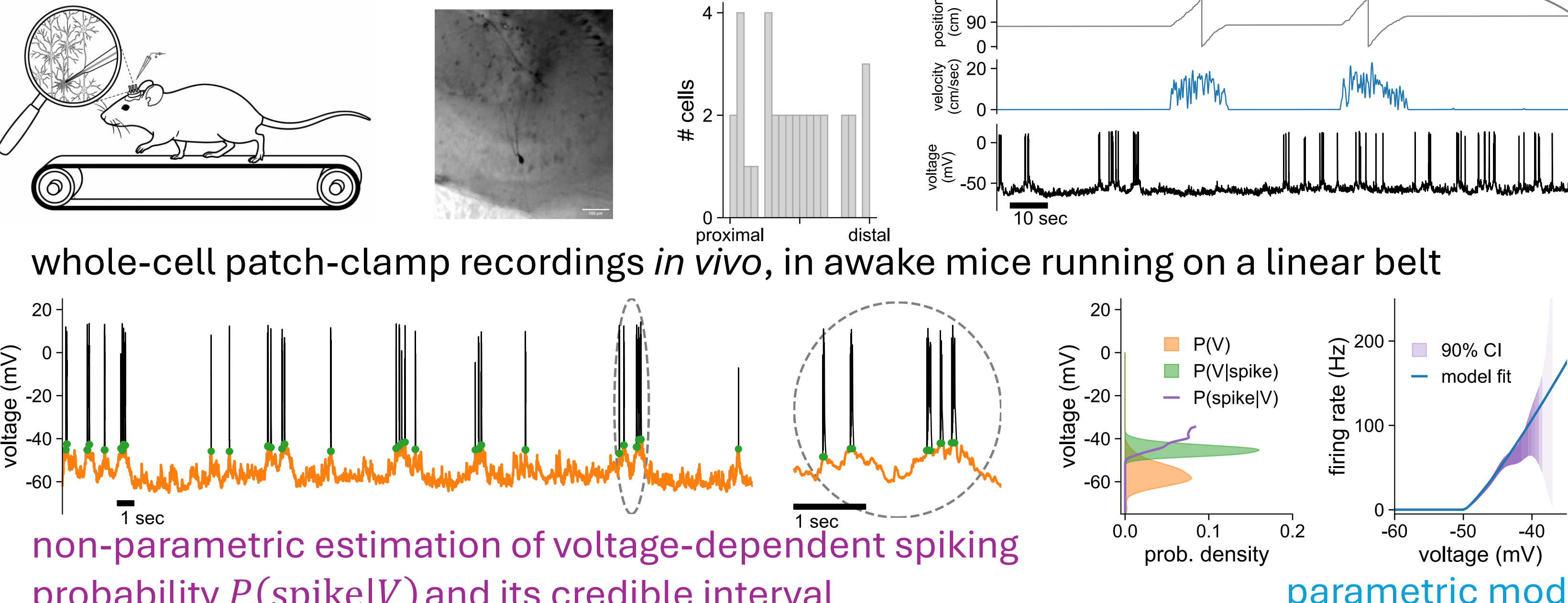
The dynamical regime of recall

predictions



- inhibition-dominated recurrent inputs
- elevated excitability (reduced threshold)
- threshold-linear activation

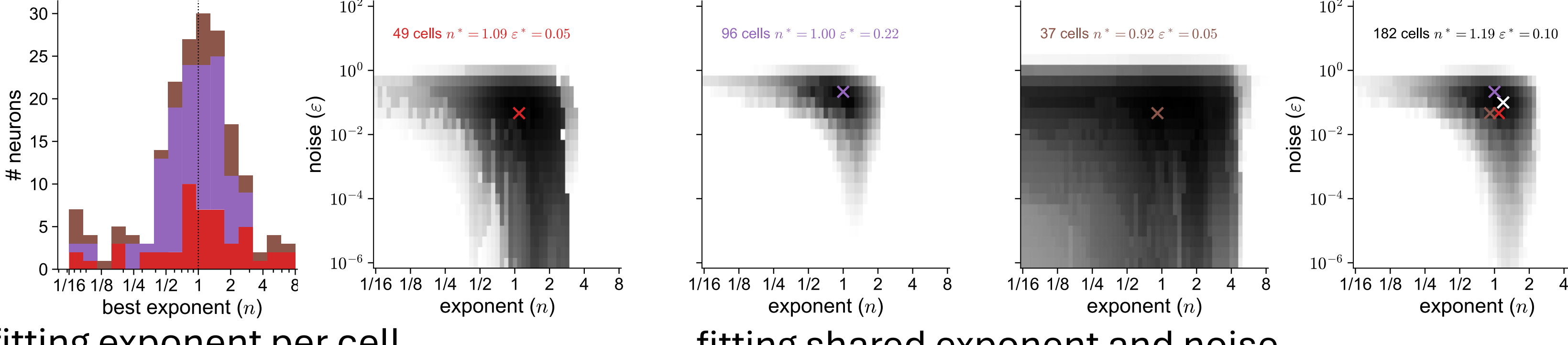
experimental results



whole-cell patch-clamp recordings *in vivo*, in awake mice running on a linear belt

non-parametric estimation of voltage-dependent spiking probability $P(\text{spike}|V)$ and its credible interval

parametric model



fitting exponent per cell

fitting shared exponent and noise

predictions status

- A net inhibitory response was observed in CA3 pyramidal neurons when activating recurrent collaterals [7], [8], per our “inhibition-dominated recurrent connectivity” prediction.
- CA3 pyramidal neurons fire spontaneously, without recurrent inputs [9], [10], per our “positive excitability” prediction.
- Analyzing new data from the Jonas lab and two previous studies [5], [6], we find that an exponent of ~1.2 best fits the data, per our “threshold-linear activation” prediction.

references

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poster

